Computer Methods and Programs in Biomedicine 175 (2019) 73–82

Contents lists available at ScienceDirect

Computer Methods and Programs in Biomedicine

journal homepage: www.elsevier.com/locate/cmpb

MRI and PET image fusion using the nonparametric density model and the theory of variable-weight

Zhe Liu[a,*], Yuqing Song[a], Victor S. Sheng[b,*], Chunyan Xu[a], Charlie Maere[a], Kaifeng Xue[a], Kai Yang[c]

[a] School of Computer Science and Telecommunication, Jiangsu University, Zhenjiang, Jiangsu Province, 212013 PR China
[b] Department of Computer Science, University of Central Arkansas, Conway, Arkansas, USA
[c] Department of Computer Science, Tongji University, Shanghai, PR China

ARTICLE INFO

Article history:
Received 30 December 2018
Revised 8 April 2019
Accepted 9 April 2019

Keywords:
MRI
Image fusion
Nonparametric density model
Multi-resolution
PET

ABSTRACT

Medical image fusion is important in the field of clinical diagnosis because it can improve the availability of information contained in images. Magnetic Resonance Imaging (MRI) provides excellent anatomical details as well as functional information on regional changes in physiology, hemodynamics, and tissue composition. In contrast, although the spatial resolution of Positron Emission Tomography (PET) provides is lower than that an MRI, PET is capable of depicting the tissue's molecular and pathological activities that are not available from MRI. Fusion of MRI and PET may allow us to combine the advantages of both imaging modalities and achieve more precise localization and characterization of abnormalities. Previous image fusion algorithms, based on the estimation theory, assume that all distortions follow Gaussian distribution and are therefore susceptible to the model mismatch problem. To overcome this mismatch problem, we propose a new image fusion method with multi-resolution and nonparametric density models (MRNDM). The RGB space registered from the source multi-modal medical images is first transformed into a generalized intensity-hue-saturation space (GIHS), and then is decomposed into the low- and high-frequency components using the non-subsampled contourlet transform (NSCT). Two different fusion rules, which are based on the nonparametric density model and the theory of variable-weight, are developed and used to fuse low- and high-frequency coefficients. The fused images are constructed by performing the inverse of the NSCT operation with all composite coefficients. Our experimental results demonstrate that the quality of images fused from PET and MRI brain images using our proposed method MRNDM is higher than that of those fused using six previous fusion methods.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

When medical imaging is required to diagnose a patient, the patient may be imaged using more than one type of imaging modalities. Different imaging tests provide different imaging features and complementary diagnostic information about the patient. Common cross-sectional clinical imaging modalities include computerized tomography (CT), Magnetic Resonance Imaging (MRI), and Positron Emission Tomography (PET). Each imaging modality has its own unique characteristics, which could be the best for the detection and analysis of *specific* clinical problems. For example, CT is excellent for the delineation of lungs, bony structures, and spatial and temporal resolution. MRI provides high-resolution, three-dimensional images with superb tissue contrast and the patient is not exposed to ionizing radiation or an iodinated contrast medium. Although PET imaging provides a lower spatial and temporal resolution than CT and MRI, it is unique and very effective in evaluating the functional and molecular characteristics of abnormalities. Although images acquired from each modality are often reviewed and interpreted independently, the fusion of images from multiple modalities may reveal diagnostic information that is difficult to visualize in each modality independently. Fused images can provide more precise localization and characterization of abnormalities [1–4].

Particularly, in this paper, we will study how to effectively fuse MRI and PET images. Existing image fusion methods can be categorized into three main groups according to their granularities, i.e., pixel-level image fusion [5], feature-level image fusion, and decision-level image fusion. Pixel-level image fusion is designed to combine multiple strictly registered images into a fused image. It can maximize the retention of complementary information in original images to a fused image. Compared with pixel-level image

* Corresponding authors.
E-mail addresses: 1000004088@ujs.edu.cn (Z. Liu), ssheng@uca.edu (V.S. Sheng).

https://doi.org/10.1016/j.cmpb.2019.04.010
0169-2607/© 2019 Elsevier B.V. All rights reserved.

fusion, the image registration requirement of feature-level image fusion is lower, but for that, detailed information inside multiple images might be lost. Compared to pixel-level and feature-level fusion, decision-level image fusion can lose much more information, since the decision is obtained through a high degree abstraction of information. In general, Pixel-level image fusion can provide more detailed information than feature-level and decision-level. Therefore, we focus on developing a novel pixel-level image fusion method in this paper for medical image fusion.

Medical image fusion methods usually belong to the pixel-level fusion group, because these approaches preserve original pixel values [6,7]. A simple pixel-level fusion method takes the average pixel values from different imaging modalities. The major drawback of pixel averaging is the reduction of image contrast. The application of the statistical estimation theory could improve the preservation of image contrast, including minimizing model distortion and mismatch. A Bayesian fusion method is an early implementation of the statistical estimation theory with the assumption that all disturbances follow a Gaussian distribution [8]. A later work from the same research group reported a region-based image fusion method based on the Expectation-Maximization approach, which allows both Gaussian and non-Gaussian disturbances to best fit observed data [9]. Another research group published a nonparametric and region-based image fusion method based on the principle of bootstrap sampling to address the model mismatch problem [10].

Recent studies on pixel-level image fusion improve the contrast and edge information on the basis of transformation [11–20], such as HIS [11,12], GIHS [13], Principal Component Analysis (PCA) [14–16] and wavelet related methods [17–19]. Among them, the method based on scale-space transformation has been rarely used alone because it may cause severe spectral distortion. The fusion method of intensity-hue-saturation (HIS) space transformation is similar to that of GIHS space transformation. In these two methods, the space of source images is transformed from Red Green Blue (i.e. RGB) to IHS or GIHS. Medical image fusion uses pseudo-color images, which are quite different from real-color images. Its fusion results often do not meet the requirements of human vision and cannot achieve the best fusion effect. This undesirable effect could be mitigated when different methods are combined. Daneshvar and Ghassemian [13] proposed an MRI and PET image fusion method by combining IHS with the retina-inspired model (RIM), which integrates the advantages of both IHS and RIM fusion methods to improve the functional and spatial information content. It provides another way to improve the fusion quality compared with other fusion methods. Li et al. [20] proposed a method of multi-scaled combination of MR and SPECT images based on variable-weight. The fusion rule is then presented with the GIHS framework and the non-subsampled contourlet transform (NSCT) for the bright visual perception, which performs better in terms of detail preservation and detail performance control. Bhatnagar et al. [21] proposed a novel framework for spatially registered multimodal medical image fusion based on NSCT. It decomposes source medical images into low- and high-frequency bands by NSCT. Low-frequency components are fused using an activity measure based on the normalized Shannon entropy, and high-frequency components are fused using the directive contrast. Compared with other methods, experimental results showed the superior performance of such a framework in terms of both subjective and objective assessment criteria.

In order to get more information and avoid the model mismatch problem, we propose a multi-resolution and nonparametric density model (called MRNDM in short). Our proposed fusion method MRNDM naturally integrates multi-resolution techniques with a nonparametric density model. The multi-resolution technique is applied to decompose original PET and MRI images into the low and high frequency sub-bands using NSCT from their GIHS space. A nonparametric density model and a maximum method based on the theory of the variable-weight theory will be used to combine corresponding low-frequency sub-bands and high-frequency sub-bands separately. Subsequently, the independently fused high- and low-frequency images are combined by the inverse NSCT transformation. Finally, we obtained the final fused image via inverse GIHS transformation. We will explain our MRNDM in details in Section 3.

The organization of this paper is as follows. In next section, we introduce related theories used in our method MRNDM, including nonparametric mixture models and the multi-resolution theory. In Section 3, we describe our integrated fusion method in detail. In Section 4, we present our experimental results by comparing with previous image fusion methods. Finally, we conclude this paper in Section 5.

2. Related theories

As we said before, our proposed fusion method MRNDM is an integration of multi-resolution techniques and a nonparametric density model. Before explaining our method MRNDM in Section 3, we first introduce the related theories, including nonparametric mixture models and the multi-resolution theory in this section.

2.1. Nonparametric mixture models

Non-parametric density estimation [22] is to estimate the overall density by directly referring to sample data, and then to describe conceptual data with the density distribution. There are various non-parametric density estimates, such as, frequency histogram estimate, kernel density estimate, orthogonal series estimate, and nearest neighbor estimate. In this paper, we utilized the orthogonal sequence non-parametric density estimation.

For a continuous unknown probability distribution f of independent and identically distributed samples $X_1, \ldots, X_N$, we could define the orthogonal polynomial density estimation [23,24] as follows.

$$f_N(x) = \sum_{-\infty}^{\infty} a_i e_i(x) \tag{1}$$

where $a_i = \frac{1}{N}\sum_{k=1}^{N} e_i(X_k)$, and $\{e_i\}_{i=-\infty}^{i=+\infty}$ is a complete orthogonal system. For practical implementation, the equation can be rewritten [23] as follows.

$$f_N(x) = \sum_{i=1}^{m} a_i e_i(x) \tag{2}$$

where $a_i = \frac{1}{N}\sum_{k=1}^{N} e_i(X_k)$, $i = 1, 2, \ldots, m$, and $\{e_i(x)\}_{i=1}^{i=m}$ is the orthogonal basis satisfied with the Hilbert orthogonal base of space $L_2([a,b])$.

2.2. Multi-resolution theories

Multi-resolution is an effective way of understanding phenomena and the nature of things. It is commonly used to perform analysis from the overall shape of objects to the internal details of things, and vice versa. Two related theories will be briefly introduced as follows.

2.2.1. RGB-IHS theory

Generalized IHS (GIHS) transformation is a widely used image fusion technique. It can exploit the complementary nature of multi-sensor image data [12,20]. Simply, GIHS fusion uses an RGB image consisting of distinct bands and transforms its distinct bands into the IHS space. In the GIHS framework, a gray-level image

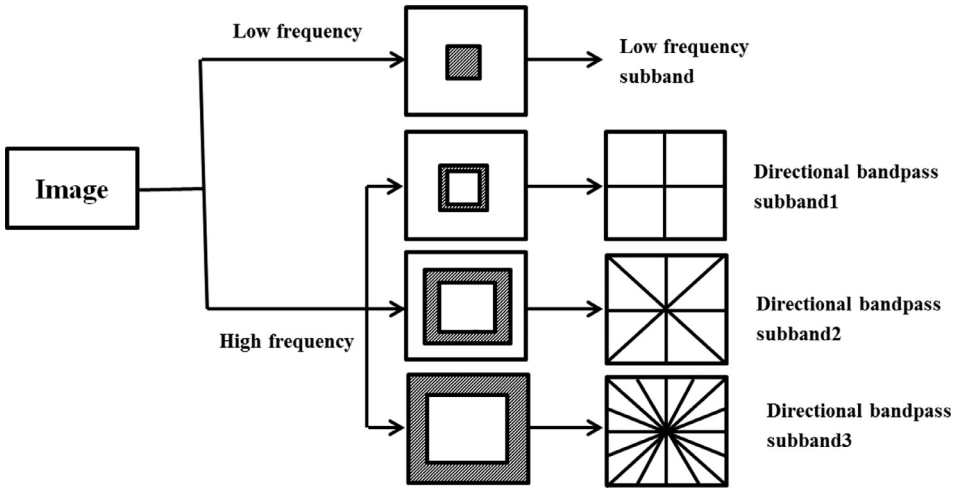


Fig. 1. A structure of non-subsampled contourlet transforms [25].

with a higher spatial resolution replaces the low-resolution intensity component in the IHS space, and then is transformed back to the original RGB space with the original Hue (H) and Saturation (S) components. The GIHS equation can be deduced through the linear transformation of the RGB-IHS as follows.

$$\begin{bmatrix} I \\ v1 \\ v2 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ -\sqrt{2}/6 & -\sqrt{2}/6 & -\sqrt{2}/6 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \tag{3}$$

where variables $v1$ and $v2$ equate the x- and y-axis in the Cartesian coordinate system while Intensity (I) from the original image represents the z-axis. In this scheme, the hue (H) and the saturation (S) could be described as follows.

$$H = \tan^{-1}\left(\frac{v2}{v1}\right) \tag{4}$$

$$S = \sqrt{v1^2 + v2^2} \tag{5}$$

where R, G, and B represent the values from a resized original multispectral image.

The inverse transformation of the GIHS equation is as follows:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & -1/\sqrt{2} & 1/\sqrt{2} \\ 1 & -1/\sqrt{2} & -1/\sqrt{2} \\ 1 & \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} I \\ v1 \\ v2 \end{bmatrix} \tag{6}$$

where R, G, and B represent the values from fused images.

2.2.2. NSCT theory

Being a shift-invariant version to the Contourlet Transform, the Nonsubsampled Contourlet Transform (NSCT) can be efficiently implemented to provide a flexible multiscale and multidirectional image decomposition [25]. NSCT is based on a non-subsampled pyramid structure and non-subsampled directional filter banks. The non-subsampled pyramid (NSP) structure ensures the multiscale property, while the non-subsampled directional filter bank (NSDFB) structure offers directionality. The initial step is to let the application of NSP to decompose an input image to a low frequency band and a high frequency band. The subsequent application of NSDFB allows a shift-invariant expansion into different directional sub-bands. Fig. 1 shows the structure of non-subsampled contourlet transforms. In each scale, the number of sub-bands could be any power of 2. NSCT is highly suitable to the image fusion process because of its following characteristics: (1) its local space and frequency domain properties are similar to the wavelet transformation: (2) its directionality property facilitates the extraction and representation of image features; and (3) in the two-step decomposition, no down-sampling process is given by its translation invariant property.

3. Nonparametric density model based on multi-resolution image fusion

Our proposed image fusion method MRNDM is based on a model of statistical image formation where distortions are permitted to best fit the data observed. Since PET images are represented in pseudo-color and treated as color images. The spatial resolution of MRI images is higher than that of PET images, but MRI images contain no color information. Our proposed fusion method MRNDM naturally integrates multi-resolution techniques with a nonparametric density model. The block architecture of our proposed method MRNDM is shown in Fig. 2.

The RGB space registered from the source multi-modal medical images is first transformed into a generalized intensity-hue-saturation space (GIHS), and then is decomposed into the low- and high-frequency components using the non-subsampled contourlet transform (NSCT).

Multi-resolution images are decomposed into the high- and low-frequency sub-bands by NSCT. The fusion scheme of high-frequency sub-band images is based on the variable-weight theory, while the low-frequency sub-band images fusion scheme is based on the expectation-maximization nonparametric density model (EMNDM). Subsequently, the independently fused high- and low-frequency images are combined by the inverse transformation of the composite coefficient representation.

3.1. Image formation model

We assume that an observed sensor image can be expressed as a linear combination of a true scene multiplied by the sensor selectivity factor and the random distortion [11]:

$$z_i(l) = \beta_i s(l) + \varepsilon_i(l), i = 1, \ldots, q \tag{7}$$

Where $i = 1, \ldots, q$ indexes corresponding sensors, $z_i(l)$ is an observed sensor image, $s(l)$ is the corresponding true scene (which we wish to approximate using fusion), $\beta_i \in \{-1,0,+1\}$ is the sensor selectivity factor, and $\varepsilon_i(l)$ is the nonparametric distortion. The value of the sensor selectivity factor presents three different states of a sensor visualization: an object visualized ($\beta_i = +1$), an object not visualized ($\beta_i = 0$), and an object visualized but with a polarity-reversed representation ($\beta_i = -1$).

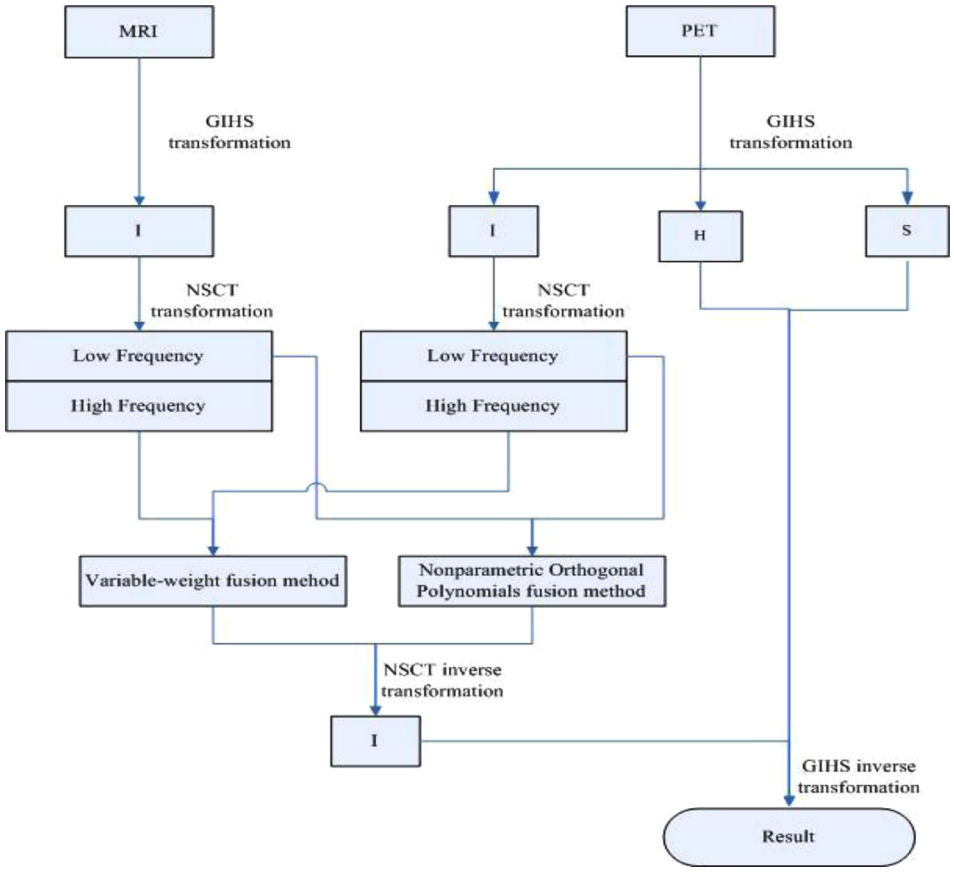


Fig. 2. The block architecture of our integrated fusion method.

The distortion that is modeled as a K-term mixture of nonparametric probability density functions can be described as follows.

$$f_{\varepsilon_i(l)}(\varepsilon_i(l)) = \sum_{k=1}^{K} w_{k,i} f_{k,K_{N_i}}(\varepsilon_i(l)) \tag{8}$$

where $f_{k,K_{N_i}}(\varepsilon_i(l)) = \sum_{j=1}^{K_{N_i}} a_{k,i,j} e_j(\varepsilon_i(l))$

While different types of orthogonal polynomials are applicable, we use the Chebyshev's second kind polynomial because of its excellent representation in image process and segmentation [26]. The second kind Chebyshev orthogonal polynomial can be described as follows.

$$e_n(x) = \frac{\sin[(n+1)\arccos x]}{\sqrt{1-x^2}} (n = 0, 1, 2, \ldots) \tag{9}$$

3.2. Image fusion using variable-weight and nonparametric density models

The external borders (e.g., the interface between soft-tissue and air, and the interface between soft-tissue and bones) of medical images are more conspicuously defined (i.e., high-frequency structures) than the boundaries of internal structures (i.e., low-frequency structures). To take advantage of this intrinsic tissue contrast, medical images can be decomposed into high- and low-frequency sub-bands. Therefore, we can apply two different fusion models for high- and low-frequency sub-bands separately. Specifically, we apply the variable-weight fusion model to the high-frequency sub-band images, and apply the EMNDM model to the low-frequency sub-band images. After these fusion are done, the high- and low-frequency images are combined by the inverse transformation of the composite coefficient representation.

3.2.1. High-frequency band fusion

In the high-frequency sub-bands, the coefficients represent the details of a source image. The high-frequency fusion theory is on how to compute the weight W_P under the limitation of Linprog; the results are computed by G_H^*, which is obtained by comparing the coefficients of MRI and PET images. The process could be defined as follows:

Step 1: Compute the extremum b at each level of high-frequency by the Linprog function.

$$\begin{cases} \forall y, \sum_{j=1}^{R} j b_{j+1} y^{j-1} \geq 0 \\ g(1) = \sum_{j=0}^{R} b_{j+1} \leq 1 \end{cases} \tag{10}$$

where R denotes the maximum of the binomial.

Step 2: Compute $g(y)$ based on the parameter b

$$g(y) = \sum_{j=0}^{R} b_{j+1} y^j \tag{11}$$

where b denotes the coefficients of high-frequency.

Step 3: Fuse the high-frequency as follows.

$$G_H^* = \begin{cases} b_H \cdot M_H(i,j), & M_H(i,j) > P_H(i,j) \\ (1-b_H) P_H(i,j), & M_H(i,j) \leq P_H(i,j) \end{cases} \tag{12}$$

$$G_H = W_P \cdot G_H^* + (I - W_P) \cdot M_H \tag{13}$$

where P denotes the high-frequency of a PET image; M denotes the high-frequency of an MRI image; W_P is the weight of the PET image; I is a unit matrix; and b_H is set as 0.5 in our later experiments.

3.2.2. Low-frequency band fusion

According to the image formation model Eq. (7) and nonparametric mixture models Eq. (8), the expectation maximization based nonparametric density model (EMNDM) is used to estimate the parameters of the models and obtain the fusion results. The algorithm of low-frequency band fusion has four steps as follows.

(1) Initialization step: The initial estimation of the true scene *s(l)* can be set as the weighted average of the sensor images. That is,

$$s(l) = \sum_{i=1}^{q} w_i z_i(l), l = 1, \ldots, L \tag{14}$$

where $w_i = 1/q$. To simplify the initialization, we assume that the object appears in each sensor image. That is, $\beta_i = 1$, $i = 1, \ldots, q$. The initial parameters w_i can be extracted by k-mean, and the class k number is assumed to be fixed.

(2) Expectation step: In this step, the conditional distribution $g_{k,i,\ l}[z_i(l)]$ is estimated as follows.

$$g_{k,i,l}(z_i(l)) = \frac{w_{k,i} f_{k,K_{N_i}}(z_i(l) - \beta_i(l)s(l))}{\sum_{k=1}^{K} w_{k,i} f_{k,K_{N_i}}(z_i(l) - \beta_i(l)s(l))} \tag{15}$$

where $i = 1, \ldots, q$ and $l = 1, \ldots, L$

(3) Maximization step: From the initial estimates, several parameters are updated (β_i', $s'(l)$, $w'_{k,i}$, $a'_{k,i,j}$) to satisfy the maximization conditions,

β'_i is selected to have the value from a set of values $\{-1, 0, 1\}$ that maximizes the following formula.

$$Q = \sum_{i=1}^{q} \sum_{l=1}^{L} \sum_{k=1}^{K} \left[\ln f_{k,K_{N_i}}\left(z_i(l) - \beta_i'(l)s(l)\right) \right] g_{k,i,l}(z_i(l)) \tag{16}$$

With the updated β'_i, $g_{k,il}(z_i(l))$ is recalculated. Subsequently, the true scene *s(l)* is recalculated:

$$s'(l) = \frac{\sum_{i=1}^{q} \sum_{k=1}^{K} z_i(l) \beta_i' g_{k,i,l}(z_i(l)) \sum_{j=1}^{K_{N_i}} j^2 a_{k,i,j}}{\sum_{i=1}^{q} \sum_{k=1}^{K} \beta_i' g_{k,i,l}(z_i(l)) \sum_{j=1}^{K_{N_i}} j^2 a_{k,i,j}}, l = 1, \ldots, L \tag{17}$$

The updated *s'(l)* and β'_i are used to recalculate the distortion parameters$w_{k,i}$and $a_{k,i,\ j}$ as follows.

$$w'_{k,i} = \frac{1}{L} \sum_{i=1}^{L} g_{k,i,l}(z_i(l)), \quad k = 1, \ldots K; \quad i = 1, \ldots, q \tag{18}$$

$$a'_{k,i,j} = \frac{\sum_{l=1}^{L} e_j\left(z_i(l) - \beta'_i s'(l)\right) g_{k,i,l}(z_i(l))}{\sum_{l=1}^{L} g_{k,i,l}(z_i(l))}, k = 1, \ldots K, i = 1, \ldots, q \tag{19}$$

where e_j is the orthogonal basis.

The Expectation and Maximization steps are iteratively processed until the parameters *s'(l)*, β'_i, $w_{k,i}$ and $a_{k,i,j}$ converge to the estimates of maximum-likelihood. The estimation *s(l)* $(l = 1, \ldots, L)$ corresponds to the result of the low-frequency band fusion.

4. Experiments

The test data consisted of the PET and MRI images of brains. In this paper, the PET images are PET-FDG (positron emission tomography with Fluorine-18 Deoxyglucose). The conventional images of PET are based on the distribution of radioactivity in different regions of the body, which means that the pseudo color is more in line with the presentation. All of the images are downloaded from the Harvard database [27]. The images of the brains could be classified into three groups (i.e., Normal, Alzheimer's disease, Neoplastic disease). Every group contains 50 cases. The spatial resolution of the PET images and MRI images is co-calibrated to 256×256 pixels. We will conduct experiments to investigate the performance of our proposed method MRNDM by comparing with six popular fusion methods, i.e., PCA [15], DWT [28], GIHS [19], Simple Average [30], NSCT [31], and EM based on Gaussian [10]. In our experiments, we followed previous research to set the parameters of each corresponding method. Some details are as follows.

PCA: The number of principal components in our experiments is set as 6. Besides, the two weighting coefficients (i.e., ω_1 and ω_2) in the optimal weighted fusion strategy are set to 0.5 and 0.5.

DWT: We used the Daubechies wavelet (dbN, $N = 4$, denoted as db4) and the number of decomposition levels is three.

GIHS: To be consistent with DWT, we also used the Daubechies wavelet (db4) and the number of decomposition levels is three.

NSCT: source images are decomposed into two levels with eight and four directional sub-bands in the first and the second decomposition level respectively.

EM-Gaussian: The image fusion method with the expectation-maximization (EM) algorithm uses a set of iterative equations to estimate the model parameters and to produce the fused image (using the final scene estimate). In this method, we set the window size $L = 5 \times 5$ and normalized the image data to [0,1].

4.1. Evaluation metrics

The wide use of multi-sensor and multi-modality medical images led to various fusion imaging processing techniques and algorithms. It is critical to select appropriate evaluation metrics to compare the performance of different fusion techniques [29]. We will investigate the performance of our proposed method MRNDM by comparing with the popular six fusion methods qualitatively and quantitatively. The qualitative evaluation will be performed by an expert radiologist who reviewed fused images from each method and compared with the original PET and MRI images. The fidelity of the image color scales, edge conspicuity and spatial resolutions between the fused and original images are qualitatively assessed. To help the expert radiologist make fair qualitative evaluation, the fusion methods are blinded.

For the quantitative assessment, six popular fusion image quality metrics (i.e., entropy, average cross entropy, average gradient, edge intensity, and mutual information (MI), and Q_{abf}) are chosen to assess the performance of all seven fusion methods.

The entropy represents the expected value of the information contained in images [30], defined as follows:

$$\mathrm{H} = -\sum_{\mathrm{i}=0}^{\mathrm{L}-1} P_i \cdot \log(P_i) \tag{20}$$

where L denotes the gray level of an image and P_i is the probability of pixel gray value in the image.

To evaluate the difference between the two images [31], the cross entropy are defined as follows:

$$CE_{A,F} = \sum_{i=0}^{L-1} P_{A_i} \log \frac{P_{A_i}}{P_{F_i}} \tag{21}$$

$$CE_{B,F} = \sum_{i=0}^{L-1} P_{B_i} \log \frac{P_{A_i}}{P_{F_i}} \tag{22}$$

where P_A and P_B are source images and P_F is the fused image.

In particular, the average cross entropy (ACE) assesses the difference between the original source and the fused images. Low ACE implies a low difference (or a high similarity) between the original source and the fused images.

$$ACE = \frac{CE_{A,F} + CE_{B,F}}{2} \tag{23}$$

Average gradient could reflect the clarity of a fused image [32]. It can also be used to measure the fused image's spatial resolution. The average gradient increases with the spatial resolution, as shown in the following equation:

$$\nabla\bar{g} = \frac{1}{M\times N}\sum_{i=0}^{L-1}\sum_{j=0}^{L-1}\sqrt{\left(\Delta I_x^2 + \Delta I_y^2\right)/2} \tag{24}$$

where ΔI_x^2 and ΔI_y^2 are the difference in the direction of x and y axis, and $M \times N$ is the size of the image.

$Q^{AB/F}$ is a measure of the similarity between the edges transferred in the fusion process [33], defined as follows.

$$Q^{AB/F} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} Q^{AF}(i,j)\lambda^A(i,j) + Q^{BF}(i,j)\lambda^B(i,j)}{\sum_{i=1}^{M}\sum_{j=1}^{N}\lambda^A(i,j) + \lambda^B(i,j)} \tag{25}$$

where A and B represent the two source images while F corresponds to the fused image. $Q^{AF}(i,j)$ and $Q^{BF}(i,j)$ are the edge preservation values. $\lambda^A(i,j)$ and $\lambda^B(i,j)$ are the weights.

Mutual information (MI) determines the statistical dependence or information redundancy between two random variables [32]. To estimate the statistical dependency between the original source and fused images, MI can be defined as follows.

$$MI((A,B);F) = \sum_{i=0}^{L-1}\sum_{j=0}^{L-1}\sum_{l=0}^{L-1} P_{abf}(i,j,l)\cdot\log_2\left(\frac{P_{abf}(i,j,l)}{P_{ab}(i,j)\cdot P_{k_f}(l)}\right) \tag{26}$$

where A and B represent the two source images while F corresponds to the fused image. To evaluate the relative degree of performance among the fusion methods in each image quality metrics, we rank our proposed method MRNDM and the six popular fusion methods according to the relative values of the performance measures in terms of each evaluation metric (i.e., 1 = best; 7 = worst). The average rank in terms of the six image quality metrics will be calculated for each method.

4.2. Experimental results

We investigated the performance of our proposed method MRNDM by comparing with the popular six fusion methods qualitatively and quantitatively on three categories of PET-MR images, i.e., normal, Alzheimer's, and neoplastic brain images. We will show the qualitative evaluation first, and then show the quantitative assessment in terms of six popular fusion image quality metrics (i.e., entropy, average cross entropy, average gradient, edge intensity, and mutual information (MI), and Q_{abf}).

4.2.1. Qualitative assessment

Fig. 3 shows normal original brain images of PET and MRI and their fusion images after processing. According to the qualitative evaluation by the expert radiologist, the color scale of the original

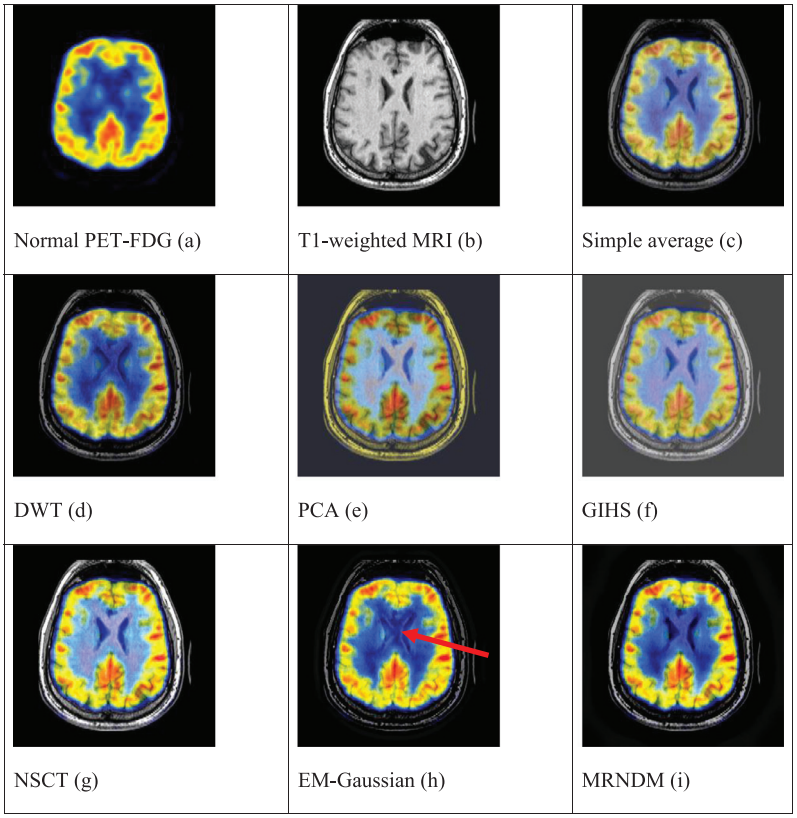


Fig. 3. Normal PET-FDG (a) and T1-weighted MRI (b), and their fusion results using different methods: simple average (c), DWT (d), PCA (e), GIHS (f), NSCT (g), EM-Gaussian (h) and our proposed method MRNDM (i). A red arrow in (h) indicates artifactual curvilinear signals within the ventricles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

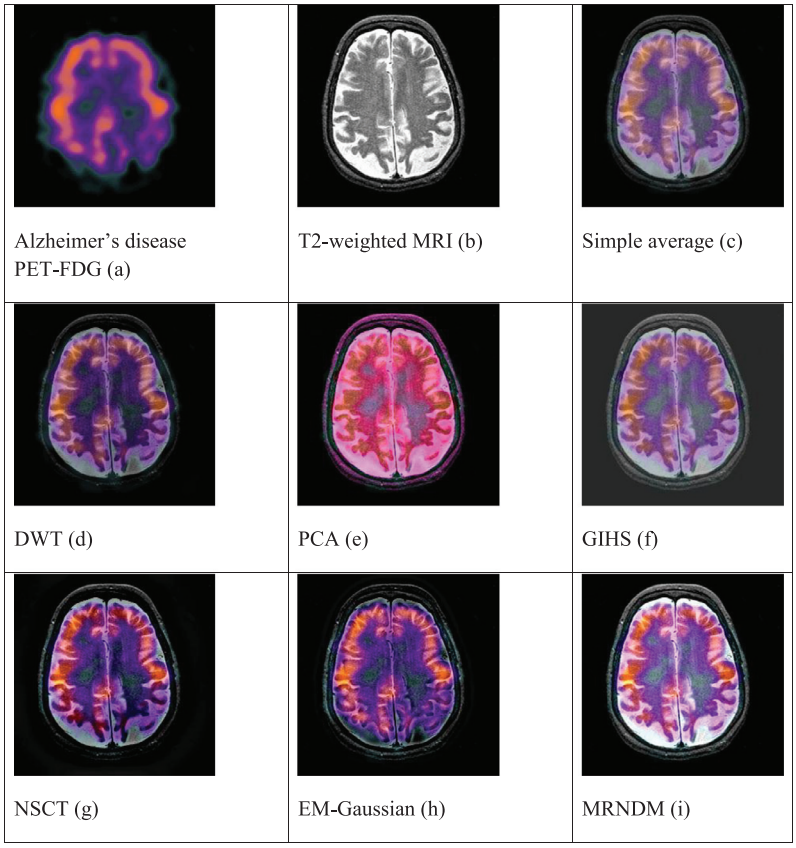


Fig. 4. Alzheimer's disease PET-FDG (a) and T2-weighted MRI (b), and their fusion results using different methods: simple average (c), DWT (d), PCA (e), GIHS (f), NSCT (g), EM-Gaussian (h) and our proposed method MRNDM (i). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

PET image is perceptually better reproduced on the images fused by the EM-Gaussian and our proposed method MRNDM than on the images fused by the other methods. The spatial resolution of the original MRI appears well preserved in all fused images except for the one fused by the EM-Gaussian method. The fused image by the EM-Gaussian method contains artifactual curvilinear signals in its ventricles.

Fig. 4 shows Alzheimer's disease original brain images of PET and MRI and their fusion images after processing. The color scales of the images fused by NSCT, EM-Gaussian and our proposed method MRNDM are relatively consistent with that of the original PET image, while the color scales of the images fused by the other methods are not preserved well. Regarding the tissue contrast and MRI spatial resolution, the image fused by our proposed method MRNDM appears the most consistent with the original MRI image.

Fig. 5 shows brain PET and MRI images with a tumor (a large astrocytoma in the right parietal-occipital region) and the fusion images after processing. The tumor presents with a high PET metabolic activity and is enhanced with MR contrast medium indicating active neoplastic process. The color scales of the images fused by NSCT and our proposed method MRNDM are relatively consistent with that of the original PET image, while the color scales of the images fused by the other methods are not consistent. Compared to other methods, the MRI tissue contrast and spatial resolution appears well preserved in the images fused by NSCT and our proposed method MRNDM.

4.2.2. Quantitative assessment

The quantitative performance of our proposed method MRNDM and other six popular fusion methods in terms of the six image quality evaluation metrics and the average rank for each fusion method are shown in Tables 1–3 on normal, Alzheimer's, and neoplastic brain images, respectively. Note that a higher value corresponds to a higher quality when the evaluation metric is entropy, edge intensity, average gradient, $Q^{AB/F}$, and mutual information. However, a lower value corresponds to a lower quality when cross entropy is the evaluation metric.

From these tables, we can see that our MRNDM performs the best on the three categories of images (i.e., normal, Alzheimer's, and neoplastic brain images) in terms of the six evaluation metrics, and its average ranks are always better than all other six popular fusion methods (i.e., Simple Average, DWT, PCA, GIHS, NSCT, EM based on Gaussian).

MRNDM performs the best in terms of the mutual information metric on all three categories of images. This implies that MRNDM has the lowest loss of spatial information among the seven fusion methods. Besides, MRNDM performs very well in terms of both entropy and cross entropy metrics (MRNDM performs the best in terms of cross entropy on the normal images, the best in terms of entropy and cross entropy on the Alzheimer's images, the best in terms of entropy and the second in terms of cross entropy on neoplastic brain images). This suggests that MRNDM has an excellent retention of information with small difference between original images and fused ones. MRNDM performs the best on both the normal and the Alzheimer's images and the second on the neoplastic brain images in terms of the average gradient metric. This indicates that MRNDM has a high degree of preserving spatial resolution characteristics of original images. Finally, MRNDM performs the best on both the Alzheimer's images and the neoplastic brain images and the second on the normal

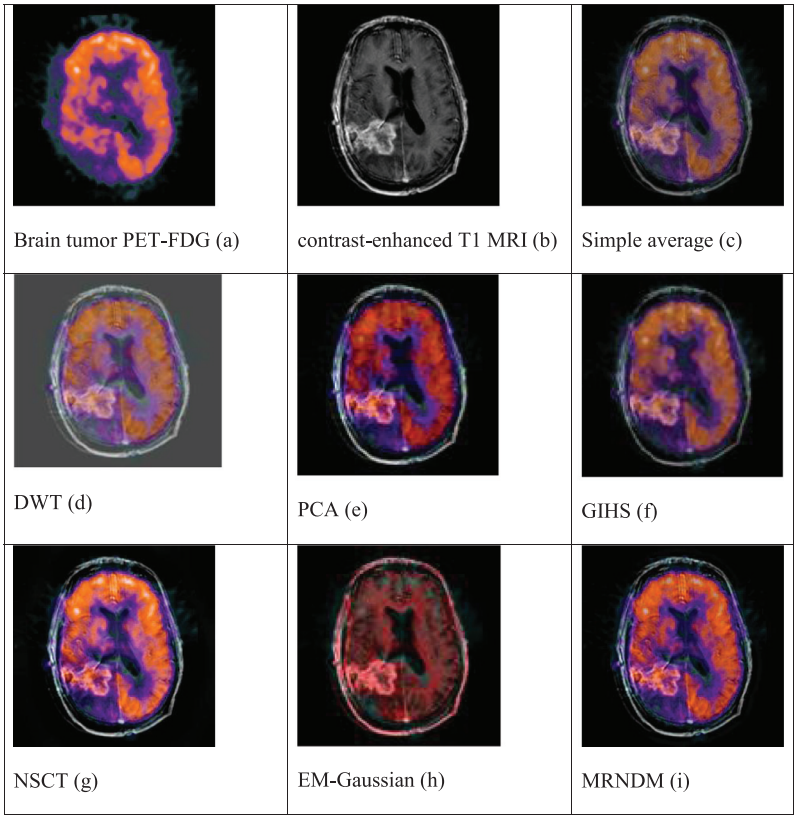


Fig. 5. Brain tumor PET-FDG (a) and contrast-enhanced T1 MRI (b), and their fusion results using different methods: simple average (c), DWT (d), PCA (e), GIHS (f), NSCT (g), EM-Gaussian (h) and our proposed method MRNDM (i). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Evaluation results of different fusion methods on normal brain PET-MRI images. The best performance is highlighted in **bold**, and the second highest is highlighted in *italic*.

Methods	Entropy	Cross entropy	Edge intensity	Average gradient	$Q^{AB/F}$	Mutual information	Average rank
Simple Average	**3.3639**	*0.3020*	58.8709	5.6045	0.3603	3.0526	3.57
DWT	2.9798	0.3352	63.9206	6.0709	0.4814	2.9861	4.00
PCA	3.1443	9.4839	58.1112	5.5404	**0.5781**	3.5629	3.71
GIHS	3.2251	10.6186	55.5526	5.2987	0.4883	3.4077	4.57
NSCT	*3.3108*	0.4062	**90.5905**	*8.1314*	0.5124	2.9404	*2.71*
EM-Gaussian	2.9735	9.4506	80.5602	7.6937	0.4336	*3.6289*	3.71
MRNDM	3.1178	**0.1558**	*87.9956*	**9.1663**	*0.5382*	**3.9135**	**1.71**

Table 2
Evaluation results of different fusion methods on Alzheimer's disease brain PET-MRI images. The best performance is highlighted in **bold**, and the second highest is highlighted in *italic*.

Methods	Entropy	Cross entropy	Edge intensity	Average gradient	$Q^{AB/F}$	Mutual information	Average rank
Simple Average	1.9974	0.7258	49.1843	4.9379	0.3237	1.6192	5.29
DWT	1.8143	3.6521	54.3468	5.0361	0.3624	0.0197	5.57
PCA	2.0998	0.7311	56.0757	5.7861	0.3914	1.7104	4.29
GIHS	2.1059	0.7266	57.1346	5.9094	0.3871	1.7106	3.71
NSCT	2.1304	0.7249	57.8051	5.9870	0.4034	1.7199	2.57
EM-Gaussian	*2.1663*	*0.7047*	*58.9902*	*6.1205*	*0.4222*	*1.7248*	*1.57*
MRNDM	**2.1743**	**0.7073**	**59.2081**	**6.1348**	**0.4365**	**1.7299**	**1.00**

Table 3
Evaluation results of different fusion methods on tumor brain PET-MRI images. The best performance is highlighted in **bold**, and the second highest is highlighted in *italic*.

Methods	Entropy	Cross entropy	Edge intensity	Average gradient	$Q^{AB/F}$	Mutual information	Average rank
Simple Average	4.2743	1.1815	42.7071	4.2514	0.3725	1.4607	5.29
DWT	4.1015	1.1576	44.1014	4.3756	0. 3935	1.1965	5.43
PCA	4.3257	1.1997	67.0685	6.7403	*0.5781*	1.6243	3.43
GIHS	3.9819	0.6186	46.0814	4.6118	0.4796	1.5790	4.43
NSCT	4.6612	0.3605	70.9601	**7.4674**	0.5429	1.6891	2.43
EM-Gaussian	*4.6672*	**0.3396**	*71.9532*	7.3217	0.5771	*1.7373*	*1.86*
MRNDM	**4.7103**	*0. 3537*	**72.9457**	*7.3318*	**0.6547**	**1. 7825**	**1.14**

brain images in terms of both $Q^{AB/F}$ and the edge intensity metric. This indicates that MRNDM can retain edge information of original images after fusion.

5. Conclusions and discussions

Since previous image fusion algorithms are based on the estimation theory and assume all distortions follow Gaussian distribution, they are susceptible to the problem of model mismatch. To overcome this problem, we proposed a new image fusion method based on multi-resolution and the nonparametric density model (EMNDM) to fuse PET and MRI images. The multi-resolution technique was used to decompose images into the low and high frequency sub-bands with NSCT from the GIHS space. We used EMNDM and the maximum method based on variable-weight theory to combine the low-frequency band and the high-frequency band separately. The low-frequency fused band is further combined with the high-frequency band using inverse NSCT transformation. Finally, we obtained the final fused image via inverse GIHS transformation.

We investigated the performance of our method MRNDM by comparing with six different fusion methods (i.e., Simple Average, DWT, PCA, GIHS, NSCT, EM based on Gaussian) qualitatively and quantitatively. Our experimental results showed that our proposed image fusion method MRNDM results in the highest overall image quality after fusing PET and MRI brain images. In terms of the average ranking, MRNDM always performs the best on the three categories of brain images (normal, Alzheimer's, and neoplastic brain images). EM based on Gaussian performs the second best on both the Alzheimer's and the neoplastic brain images, followed by NSCT. NSCT performs the second best on the normal brain images, followed by EM based on Gaussian. In summary, MRNDM performs better in terms of preserving the fidelity of the image color scales, edge conspicuity and spatial resolutions, comparing with the six different fusion methods. It consistently performs the best on the three different types of the PET-MR brain images.

There is a limitation in our study. The PET and MR brain images used in our experiments are available only in the axial plane. The PET images do not contain parametric maps. In a typical clinical setting, PET and MR images are displayed and reviewed in multiplanes including PET parametric maps. Given this limited availability of images, we could not investigate its performance in multi-planar or multi-parametric representations. We need a further validation test on multi-planar and multi-parametric images with diverse pathologies to develop a clinical tool with our method MRNDM.

Conflict of interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgments

The authors would like to thank Dr. Zhu of the Department of Medical at the University of Jiangsu. This work was supported by the National Natural Science Foundation of China (61772242, 61728205, 61572239, 61402204), the Natural Science Foundation of Jiangsu Province (BK20130529), the Research Fund for Advanced Talents of Jiangsu University (14JDG141), the Science and Technology Project of Zhenjiang City (SH20140110), the Special Software Development Foundation of Zhenjiang City (Project No. 201322), and the Science and Technology Support Foundation of Zhenjiang City (Industrial) (GY2014013).

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cmpb.2019.04.010.

References

[1] A.P. James, B.V. Dasarathy, Medical image fusion: a survey of the state of the art, Inf. Fusion 19 (2014) 4–19.
[2] S. Park, J. Jang, J. Kim J, et al., Real-time triple-modal photoacoustic, ultrasound, and magnetic resonance fusion imaging of humans, IEEE Trans. Med. Imaging (2017) pp (99): 1-1.
[3] X. Liu, W. Mei, H. Du, Structure tensor and nonsubsampled shearlet transform based algorithm for CT and MRI image fusion, Neurocomputing 235 (2017) 131–139.
[4] Q. Shile, D.C. Vince, G.M. van Erp Theo, et al., Multimodal fusion with reference: searching for joint neuromarkers of working memory deficits in schizophrenia, IEEE Trans. Med. Imaging, 37 (1) (2018) 93–105.
[5] S. Li, X. Kang, L. Fang, et al., Pixel-level image fusion: a survey of the state of the art, Inf. Fusion 33 (2017) 100–112.
[6] G. Bhatnagar, Q.M.J. Wu, Z. Liu, Human visual system inspired multi-modal medical image fusion framework, Expert Syst. Appl. 40 (5) (2013) 1708–1720.
[7] L. Yang, B.L. Guo, W. Ni, Multimodality medical image fusion based on multiscale geometric analysis of contourlet transform, Neurocomputing 72 (1–3) (2008) 203–211.
[8] R.K. Sharma, et al., Bayesian sensor image fusion using local linear generative models, Opt. Eng. 40 (7) (2001) 1364–1376.
[9] J. Yang, R.S. Blum, A statistical signal processing approach to image fusion for concealed weapon detection, in: IEEE Int. Conf. Image Processing, Rochester, NY, 2002, pp. 513–516.
[10] J. Yang, R.S. Blum, A region-based image fusion method using the expectation–maximization algorithm, in: Inf. Sci. Syst., 40th Annu. Conf., 2006, pp. 468–473.
[11] M. Zribi, Non-parametric and region-based image fusion with Bootstrap sampling, Inf. Fusion 11 (2010) 85–94.
[12] L.F. Zhang, Z.Y. Li, J. Liu, C.Y. Li, Improved algorithm of adaptive DR image denoising based on fast Curvelet transform and anisotropic median-diffusion filtering, Multimedia Syst. 12 (43) (2014) 581–584.
[13] S. Daneshvar, H. Ghassemian, MRI and PET image fusion by combining IHS and retina-inspired models, Inf. Fusion 11 (2010) 114–123.
[14] L. Dong, Q. Yang, H. Wu, et al., High quality multi-spectral and panchromatic image fusion technologies based on Curvelet transform, Neurocomputing 159 (2015) 268–274.
[15] Y.S. Li, J.H. Qu, W.Q. Dong, Y.X. Zheng, Hyperspectral pansharpening via improved PCA approach and optimal weighted fusion strategy, Neurocomputing 315 (13) (2018) 371–380.
[16] Desale, R. Pandit, S.V. Verma, Study and analysis of PCA, DCT & DWT based image fusion techniques, signal process, image process, Pattern Recognit. (ICSIPR), 2013 Int. Conf. IEEE, 2013.
[17] H. Wei, M. Viallon, B. Delattre, K. Moulin, F. Yang, P. Croisille, Y. Zhu, Free-breathing diffusion tensor imaging and tractography of the human heart in healthy volunteers using wavelet-based image fusion, Med. Imaging IEEE Trans. 34 (1) (2015) 306–316.
[18] G. Pajares, J.M. de la Cruz, A wavelet-based image fusion tutorial, Pattern Recognit. 37 (9) (2004) 1855–1872.
[19] R. Singh, A. Khare, Fusion of multimodal medical images using Daubechies complex wavelet transform – a multiresolution approach, Inf. Fusion 19 (2014) 49–60.
[20] T. Li, Y. Wang, Multiscaled combination of MRI and SPECT images in neuroimaging: a simplex method based variable-weight fusion, Comput. Meth. Programs Biomed. 105 (1) (2012) 31–39.
[21] G. Bhatnagar, Q.M.J. Wu, Z. Liu, A new contrast based multimodal medical image fusion framework, Neurocomputing 157 (2015) 143–152.
[22] R. Sheikhpour, M.A. Sarram, M.A.Z. Chahooki, et al., A kernelized non-parametric classifier based on feature ranking in anisotropic Gaussian kernel, Neurocomputing 267 (2017) 545–555.
[23] N.N. Cencov, Evaluation of an unknown distribution density from observations, Sov. Math. 3 (1962) 559–1562.
[24] K.R. Gehringer, R.A. Redner, Nonparametric Probability density estimation using normalized B-splines, Commun. Stat.-Simul. Comput. 21 (3) (1992) 849–878.
[25] A.L. Da Cunha, J. Zhou, M.N. Do, The nonsubsampled contourlet transform: theory, design, and applications, IEEE Trans. Image Process. 15 (10) (2006) 3089–3100.
[26] Z. Liu, Y. Song, J. Chen, Image segmentation based on non-parametric mixture models of Chebyshev orthogonal polynomials of the second kind, Comput. Res. Dev. 48 (11) (2011) 2008–2014.
[27] Neuroimaging Primer - Harvard Medical School Lecture Notes: Introduction to Neuroimaging. http://www.med.harvard.edu/AANLIB/home.html.
[28] C.K. Solanki, N.M. Patel, Pixel based and wavelet based image fusion methods with their comparative study, Natl. Conf. Recent Trends Eng. Technol. (2011) 13–14.
[29] Z. Liu, E. Blasch, G. Bhatnagar, et al., Fusing synergistic information from multi-sensor images: an overview from implementation to performance assessment, Inf. Fusion 42 (2018) 27–145.

[30] O. Prakash, R. Srivastava, A. Khare, Biorthogonal wavelet transform based image fusion using absolute maximum fusion rule, In: Inf. Commun. Technol. (ICT), 2013 IEEE Conf., pp. 577–582.
[31] P. Ganasala, V. Kumar, CT and MR image fusion scheme in nonsubsampled contourlet transform domain, J. Digit. Imaging 27 (3) (2014) 407–418.
[32] B.S. Kumar, Image fusion based on pixel significance using cross bilateral filter, Signal Image Video Process 9 (5) (2015) 1193–1204.
[33] G. Bhatnagar, Q.M. Wu, Z. Liu, Directive contrast based multimodal medical image fusion in NSCT domain, Multimed. IEEE Trans. 15 (5) (2013) 1014–1024.